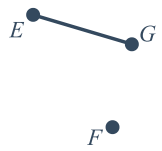
Line segments and distance

1



2 $\overline{AC}$

3 $\overline{XW}$

4 $\overline{AE}$

5 $\overline{DF}$

6

a 5 units

b 5 units

7

a 4 units

b 1 unit

c $c = \sqrt{17}$

8

a 2 units

b 1 unit

9 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

10 Distance $= \sqrt{(-5 - 4)^2 + (10 - (-5))^2}$

11 10.00

12

a 15 units

b $\sqrt{73}$ units

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13 $\sqrt{34}$ units

14

a $\sqrt{82}$ units

b $\sqrt{26}$ units

c N

15

a $\sqrt{50}$ units

b 5 units

c M

16

a $\sqrt{50}$ units

b $\sqrt{2}$ units

c $5 : 1$

17

a $CE = 28$

b $GH = 15$

18

a $QS = 63$

b $PQ = 24$

19

a $x = 3$

b 25

20 $JT = 40$

21

a

i $x = 3$

ii 6

b

i $x = 3$

ii 3

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Additional questions

22

a

i $AB = 3$ units

ii $BC = 4$ units

iii $AC = 5$ units

b

i $AB = 4$ units

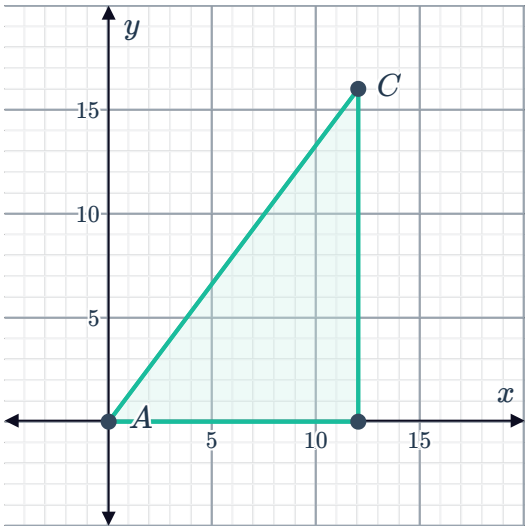
ii $BC = 7$ units

iii $AC = 8.06$ units

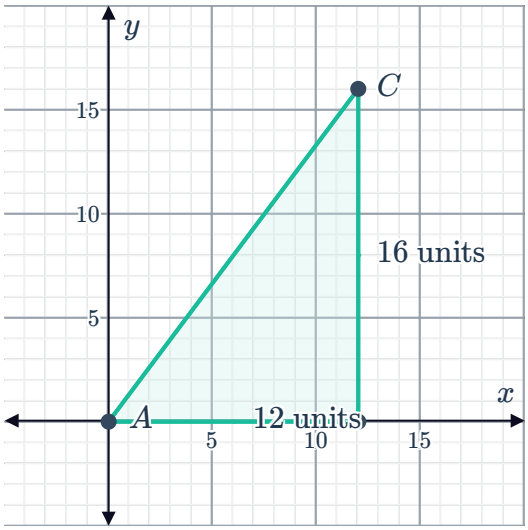
23

a

i



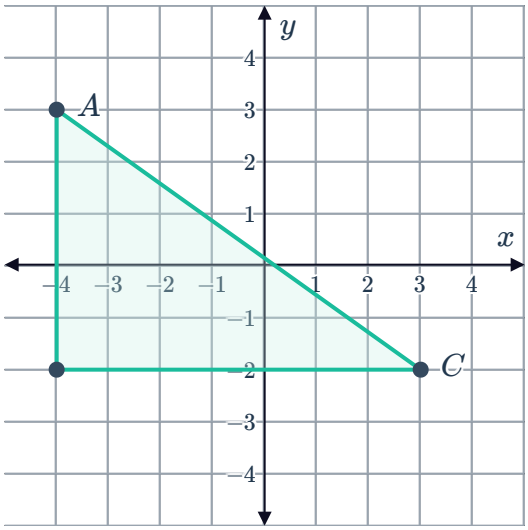
ii



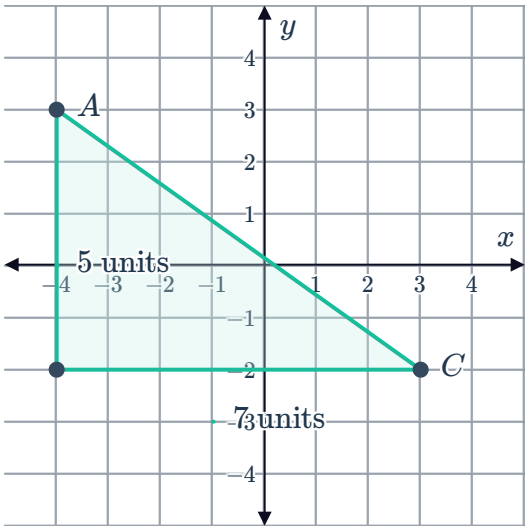
iii $AC = 20$ units

b

i



ii



iii $AC = 8.6$ units

24

a $AC = 7.81$ units

b $AC = 11.40$ units

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25

a 10 units

b 15 units

c 8.54 units

d 5.83 units

26

a $PM = 9.90$ units

b $PN = 11.18$ units

c N

27 17 units

28

a $AC = 14.14$ units

b $BD = 8.49$ units

29

a Culmer Station is the closest to their meeting spot.

b When actually walking, we need to consider traffic lights, side walks and what is accessible. Depending on which direction they are catching the train may also impact their choice.

c If they need to stop at the Winn-Dixie, then the Civic Center Station would have the shortest trip distance.

30

a There are two possible coordinates.

b $M\left(-2, 9\right)$ and $M\left(-2, 3\right)$

c If we were not told the line segment was vertical, there would be an infinite number of answers.

31 The distance formula comes from the Pythagorean theorem. $\left(x_2 - x_1\right)$ is the calculationg for the length of the line segment which is the horizontal component of the right triangle we can create. $\left(y_2 - y_1\right)$ is the calculationg for the length of the line segment which is the vertical component of the right triangle we can create.

32 Both Dayanna and David are correct. $\left(x_2 - x_1\right)$ and $\left(x_1 - x_2\right)$ will have the same magnitude, but different signs. However, when we square them, they will both become positive, so they will give the same answer.

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33

a Fyfe made an error on Step 2. They incorrectly substituted $(x_2 - y_2)$ and $(y_1 - x_1)$, but this calculation does not connect to the Pythagorean theorem to make a right triangle and so does not find the distance.

b

1	$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Distance formula
2	$= \sqrt{(3 - -4)^2 + (-5 - 7)^2}$	Substitution Property of Equality
3	$= \sqrt{(7)^2 + (-12)^2}$	Simplify
4	$= \sqrt{49 + 144}$	Simplify
5	$= \sqrt{193}$	Simplify

34

a $AC = 1$

b $AE = 4$

c $EF = 1$

d $FB = 2$

e $DB = 4$

f $AB = 7$

35

a By the segment addition postulate we know that $CD + DE = CE$ and so $CE = 28$

b By the segement addition postulate we know that $FG + GH = FH$ and so $GH = 15$

36

a $PR = 5.1 \text{ mm}$

b $AB = 51.4 \text{ cm}$

37

a $\overline{DE}$ and $\overline{EF}$

b $\overline{CF}$ and $\overline{DB}$

c In part (a) you would find two new congruent segments, $\overline{CG}$ and $\overline{DG}$, and in part (b) you would not find any additional congruent segments.

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38

a $KL = 25$

b $JT = 40$

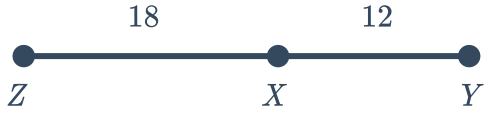
39

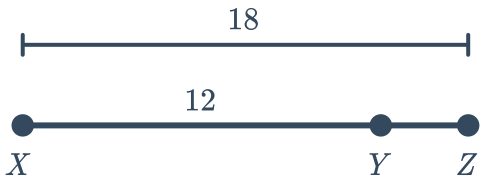
a $BD = 12$

b $CB = 3$

40 $ML = 51$

41 YZ could be 30 units in length or 6 units in length:





42 By the segment addition postulate, if points A, B, and C were collinear, then $AB + BC = AC$.
Since $AB + BC = 15$ and $AC = 12$, we know that these points cannot be collinear.

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